12. According to paragraph 5, eastern Texas is an example of a geologic region where
O oil is located but has not yet been pumped out
O experts accurately predicted the rock types and structures found there
O the volume of oil can only be guessed at
O intensive oil exploration has occurred over a long time

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Because such muds are a major source of petroleum, scientists believe that petroleum originated as living organisms.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage This question is worth 2 points.

Drag your answer choices to the spaces where they belong To remove an answer choice, click on it.

To review the passage, click VIEW TEXT

Petroleum, which includes both oil and natural gas, can be a gaseous, liquid, or semisolid substance.

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Answer Choices

Petroleum comes from organic matter that has undergone a complex series of chemical changes under the seafloor.

Although most of the petroleum formed leaks away into the ocean, some migrates from shale to sandstone or limestone, and is caught in pools.

Porous rocks made of quartz or carbonate minerals are particularly likely to house oil pools because of their strong molecular attraction with oil.

Petroleum forms best when organic matter is evenly distributed over a large area and does not exceed 8 percent of the material in the clay.

More than 60 percent of the petroleum discovered so far has been found in rocks that are less than two-and-a-half million years old.

It is difficult to estimate the total amount of petroleum in the world, but experts believe that 1,500-3,000 billion barrels will eventually be discovered.

TPO 43

[Reading]

1.The Empire of Alexander the Great

In 334 B.C. Alexander the Great took his Greek armies to the east and in only a few years completed his creation of an empire out of much of southwest Asia In the new empire, barriers to trade and the movement of peoples were removed; markets were put in touch with one another. In the next generation thousands of Greek traders and artisans would enter this wider world to seek their fortunes. Alexander's actions had several important consequences for the region occupied by the empire.

The first of these was the expansion of Greek civilization throughout the Middle East. Greek became the great international language. Towns and cities were established not only as garrisons (military posts) but as centers for the diffusion of Greek language, literature, and thought, particularly through libraries, as at Antioch (in modern Turkey) and the most famous of all, at Alexandria in Egypt, which would be the finest in the world for the next thousand years.

Second, this internationalism spelled the end of the classical Greek city-state——the unit of government in ancient Greece——and everything it stood for. Most city-states had been quite small in terms of citizenry, and this was considered to be a good thing. The focus of life was the agora, the open marketplace where assemblies could be held and where issues of the day, as well as more fundamental topics such as the purpose of government or the relationship between law and freedom, could be discussed and decisions made by individuals in person. The philosopher Plato (428-348 B.C.) felt that the ideal city-state should have about 5,000 citizens, because to the Greeks it was important that everyone in the community should know each other. In decision making, the whole body of citizens together would have the necessary knowledge in order generally to reach the right decision, even though the individual might not be particularly qualified to decide. The philosopher Aristotle (384-322 B.C.), who lived at a time when the city-state system was declining, believed that a political entity of 100,000 simply would not be able to govern itself.

This implied that the city-state was based on the idea that citizens were not specialists but had multiple interests and talents——each a so-called jack-of-all-trades who could engage in many areas of life and politics. It implied a respect for the wholeness of life and a consequent dislike of specialization. ■ It implied economic and military self-sufficiency. ■ But with the development of trade and commerce in Alexander's empire came the growth of cities; it was no longer possible to be a jack-of-all-trades. ■ One now had to specialize, and with specialization came professionalism. ■ There were getting to be too many persons to know, an easily observable community of interests was being replaced by a multiplicity of interests. The city-state was simply too "small-time."

Third, Greek philosophy was opened up to the philosophy and religion of the East At the peak of the Greek city-state, religion played an important part. Its gods—such as Zeus, father of the gods, and his wife Hera—were thought of very much as being like human beings but with superhuman abilities. Their worship was linked to the rituals connected with one's progress through life—birth, marriage, and death— and with invoking protection against danger, making prophecies, and promoting healing, rather than to any code of behavior. Nor was there much of a theory of afterlife.

Even before Alexander's time, a life spent in the service of their city-state no longer seemed ideal to Greeks The Athenian philosopher Socrates (470-399 B.C.) was the first person in Greece to propose a morality based on individual conscience rather than the demands of the state, and for this he was accused of not believing in the city's gods and so corrupting the youth, and he was condemned to death. Greek philosophy—or even a focus on conscience—might complement religion but was no substitute for it, and this made Greeks receptive to the religious systems of the Middle East, even if they never adopted them completely The combination of the religious instinct of Asia with the philosophic spirit of Greece spread across the world in the era after Alexander's death, blending the culture of the Middle East with the culture of Greece.

1. According to paragraph 1, Alexander the Great did which of the following?
O Regulated the movement and resettlement in southwest Asia of thousands of Greek people
O Opened up opportunities in new markets for traders and artisans
O Created new restrictions on trade
O Encouraged Greek citizens to choose military careers over careers in trade

2. The word "diffusion" in the passage is closest in meaning to
O adoption
O spread
O teaching
O learning

3. In paragraph 2,the author mentions the libraries at Antioch and Alexandria in order to
O provide evidence that the library was a cultural institution in the East before it spread to the West
O explain why it was important for Greek to become the great international language
O identify two of the sources of Greek cultural influence within Alexander's empire
O support the claim that the Greeks transformed Middle Eastern garrisons and military posts into cultural centers

4. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
O Assemblies were held in the agora to discuss some issues of the day, but more fundamental questions were decided by key individuals.
O In a culture where philosophical discussions were frequent, some individuals questioned the value of a life focused on the marketplace.
O Life centered around the agora, an open marketplace and site for public debate, where individuals could participate in decision making.
O The focus of individuals was on fundamental topics such as the purpose of government and the connection between law and freedom.

5. According to paragraph 3， Plato believed that the ideal city-state should be
O governed by a ruling body of about 5,000 city leaders with a total population of no more than 100,000
O led by the most qualified individual
O governed by the group of citizens with the most knowledge about the issues of the day

O small enough so that everyone would know each other

6. Why does the author mention "The philosopher Aristotle"?
O To provide additional evidence that the ancient Greeks believed that political units must be small
O To demonstrate the accuracy of philosophers’ predictions about the end of the classical Greek city-state
O To show how changes in the city-state system from the fifth to the third century B C. were reflected in the ideas of its philosophers
O To support the claim that small city-states were ideally suited to produce philosophical inquiry

7. The word "declining" in the passage is closest in meaning to
O at its best
O rapidly expanding
O first being formed
O weakening

8. According to paragraph 4, Alexander's empire was characterized by all of the following EXCEPT
O decreased need for military control
O growing professionalism
O growth of cities
O specialization in trades

9. The word ”peak” in the passage is closest in meaning to
O end
O command
O high point
O beginning

10. According to paragraph 5, religion in the Greek city-state involved
O a set of rules governing behavior
O a detailed conception of life after death
O rituals related to significant life events
O worship of gods who were not like humans

11. According to paragraph 6, what was the basis for the accusation against Socrates?
O He encouraged people to be guided by their own consciences instead of by the state.
O He stated that people had a duty to fight against the corruption of their leaders.
O He reasoned that the needs of the youth were more important than the needs of the state.
O He argued that people’s behavior should be guided by the religious systems of the Middle East.

12.The word "propose" in the passage is closest in meaning to
O suggest

O deny
O consider
O question

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Likewise, the collective decision-making process of the open marketplace was no longer practical.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your answer choices to the spaces where they belong To remove an answer choice, click on it.

To review the passage, click VIEW TEXT

Alexander the Great's creation of a vast empire had important consequences for Greece and the conquered areas of southwest Asia.

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Answer choices

Scholars from Antioch, Alexandria, and other Middle Eastern cultural centers came to Greece to study the Greek language and culture.

Increasing urbanization and the elimination of trade barriers meant the end of the Greek city-state and the creation of a much larger political and economic body.

The professionalism and specialization so prized by the ancient Greeks were replaced by a more generalized philosophy of education in the empire.

The expansion of Alexander's empire led to the diffusion of Greek language, literature, and thought throughout the Middle East.

The empire saw the birth of a new culture, merging Greek philosophical ideas with the religious spirit of Asia.

Religion played an important part in the expansion of the empire, as Alexander introduced Zeus and the other Greek gods to Asia.

2. The Origin of Petroleum

Petroleum is defined as a gaseous, liquid, and semisolid naturally occurring substance that consists chiefly of hydrocarbons (chemical compounds of carbon and hydrogen). Petroleum is therefore a term that includes both oil and natural gas. Petroleum is nearly always found in marine sedimentary rocks. In the ocean, microscopic phytoplankton (tiny floating plants) and bacteria (simple, single-celled organisms) are the principal sources of organic matter that is trapped and buried in sediment Most of the organic matter is buried in clay that is slowly converted to a fine-grained sedimentary rock known as shale. During this conversion, organic compounds are transformed to oil and natural gas.

■ Sampling on the continental shelves and along the base of the continental slopes has shown that fine muds beneath the seafloor contain up to 8 percent organic matter. ■ Two additional kinds of evidence support the hypothesis that petroleum is a product of the decomposition of organic matter: oil possesses optical properties known only in hydrocarbons derived from organic matter, and oil contains nitrogen and certain compounds believed to originate only in living matter. ■ A complex sequence of chemical reactions is involved in converting the original solid organic matter to oil and gas, and additional chemical changes may occur in the oil and gas even after they have formed. ■

It is now well established that petroleum migrates through aquifers and can become trapped in reservoirs. Petroleum migration is analogous to groundwater migration. When oil and gas are squeezed out of the shale in which they originated and enter a body of sandstone or limestone somewhere above, they migrate readily because sandstones (consisting of quartz grains) and limestones (consisting of carbonate minerals) are much more permeable than any shale. The force of molecular attraction between oil and quartz or carbonate minerals is weaker than that between water and quartz or carbonate minerals. Hence, because oil and water do not mix, water remains fastened to the quartz or carbonate grains, while oil occupies the central parts of the larger openings in the porous sandstone or limestone. Because oil is lighter than water, it tends to glide upward past the carbonate- and quartz-held water. In this way, oil becomes segregated from the water; when it encounters a trap, it can form a pool.

Most of the petroleum that forms in sediments does not find a suitable trap and eventually makes its way, along with groundwater, to the surface of the sea. It is estimated that no more than 0.1 percent of all the organic matter originally buried in a sediment is eventually trapped in an oil pool. It is not surprising, therefore, that the highest ratio of oil and gas pools to volume of sediment is found in rock no older than 2.5 million years—young enough so that little of the petroleum has leaked away—and that nearly 60 percent of all oil and gas discovered so far has been found in strata that formed in the last 65 million years This does not mean that older rocks produced less petroleum; it simply means that oil in older rocks has had a longer time in which to leak away.

How much oil is there in the world? This is an extremely controversial question. Many billions of barrels of oil have already been pumped out of the ground. A lot of additional oil has been located by drilling but is still waiting to be pumped out Possibly a great deal more oil remains to be found by drilling. Unlike coal, the volume of which can be accurately estimated, the volume of undiscovered oil can only be guessed at. Guesses involve the use of accumulated experience from a century of drilling. Knowing how much oil has been found in an intensively drilled area, such as eastern Texas, experts make estimates of probable volumes in other regions where rock types and structures are similar to those in eastern Texas. Using this approach and considering all the sedimentary basins of the world, experts estimate that somewhere between 1,500 and 3,000 billion barrels of oil will eventually be discovered.

1.According to paragraph 1, petroleum is formed in which of the following ways?
O Bacteria and tiny plants undergo a change while they are buried in clay.
O Carbon and hydrogen combine to form shale.
O Ocean rocks are converted into organic compounds.
O Oil and gas rise to the surface of sediment and are trapped in rocks.

2.The word "trapped" in the passage is closest in meaning to
O hidden
O destroyed
O caught
O found

3. All of the following are cited in paragraph 2 as evidence that petroleum is a product of the decomposition of organic matter EXCEPT
O the amount of organic matter found in layers of mud below the seafloor
O the chemical changes that occur in oil and natural gas after they have formed
O the optical properties of oil
O the fact that oil contains nitrogen and other compounds believed to be of organic origin

4. According to paragraph 2, which of the following is true of the change of solid organic material into oil and gas?
O It is more likely to occur along the base of continental slopes than on the continental shelves.
O It only takes place in areas where the seafloor contains at least 8 percent organic matter.
O It is a process that can be reversed through chemical changes that occur after the oil and gas have formed.
O It involves a complicated series of chemical reactions.

5. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
O When oil and gas are squeezed out of the rock in which they originated, it is probably because the layer of rock above them is much more permeable than shale.
O Sandstones, which are made of quartz grains, and limestones, which are made of carbonate minerals, can hold much more oil and gas than any shale can.
O When they are squeezed from the shale in which they were formed, oil and gas move easily into the much more permeable layers of sandstone or limestone above.
O Oil and gas are squeezed out of sandstones, consisting of quartz grains, and migrate readily into limestones, which consist of carbonate minerals and are much more permeable.

6. Why does the author include the information that "The force of molecular attraction between oil and quartz or carbonate minerals is weaker than that between water and quartz or carbonate minerals."?
O To help explain why petroleum behaves differently from water in bodies of sandstone and limestone
O To illustrate why petroleum migrates more rapidly through sandstone than it does through limestone
O To help explain how water and petroleum can mix in certain aquifers
O To account for the different molecular structures of oil and water

7. The word "encounters" in the passage is closest in meaning to
O meets
O forms
O escapes
O avoids

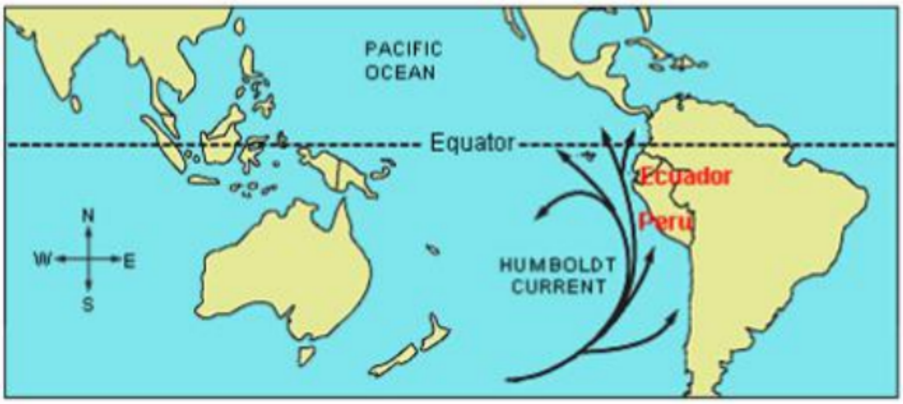
8. The word "suitable" in the passage is closest in meaning to
O noticeable
O permanent
O protected
O appropriate

9. According to paragraph 4, what happens to most of the petroleum that forms in sediments?
O It remains in underground pools.
O It is buried under organic matter.
O It rises to the surface of the ocean.
O It combines with the minerals found in groundwater.

10. Paragraph 4 supports which of the following statements about future petroleum discoveries?
O Less petroleum will be found than in the past because the ratio of petroleum pools to volume of sediment will decrease.
O Most of the petroleum will come from rocks that are less than 65 million years old.
O Petroleum that has leaked away from older rocks will be the source of most new discoveries.
O More petroleum will become available because the amount of trapped organic matter will increase.

11. The phrase "is an extremely controversial question" in the passage is closest in meaning to
O is a question of great importance
O is a question causing strong disagreement
O is a question that has existed for a long time
O is a question composed of many related parts

3.El Nirio



The cold Humboldt Current of the Pacific Ocean flows toward the equator along the coasts of Ecuador and Peru in South America. When the current approaches the equator, the westward-flowing trade winds cause nutrient-rich cold water along the coast to rise from deeper depths to more shallow ones. This upwelling of water has economic repercussions. Fishing, especially for anchovies, is a major local industry.

Every year during the months of December and January, a weak, warm countercurrent replaces the normally cold coastal waters. Without the upwelling of nutrients from below to feed the fish, fishing comes to a standstill. Fishers in this region have known the phenomenon for hundreds of years. In fact, this is the time of year they traditionally set aside to tend to their equipment and await the return of cold water. The residents of the region have given this phenomenon the name of El Nifio， ,which is Spanish for "the child," because it occurs at about the time of the celebration of birth of the Christ child.

While the warm-water countercurrent usually lasts for two months or less, there are occasions when the disruption to the normal flow lasts for many months. In these situations, water temperatures are raised not just along the coast, but for thousands of kilometers offshore. Over the last few decades, the term El Nifio has come to be used to describe these exceptionally strong episodes and not the annual event. During the past 60 years, at least ten El Nifios have been observed Not only do El Niftos affect the temperature of the equatorial Pacific, but the strongest of them impact global weather.

The processes that interact to produce an El Nifio involve conditions all across the Pacific, not just in the waters off South America. Over 60 years ago, Sir Gilbert Walker, a British scientist, discovered a connection between surface pressure readings at weather stations on the eastern and western sides of the Pacific. He noted that a rise in atmospheric pressure in the eastern Pacific is usually accompanied by a fall in pressure in the western Pacific and vice versa. He called this seesaw pattern the Southern Oscillation. It was later realized that there is a close link between El Nino and the Southern Oscillation. In fact, the link between the two is so great that they are often referred to jointly as ENSO (El Nino-Southern Oscillation).

During a typical year, the eastern Pacific has a higher pressure than the western Pacific does. This east-to-west pressure gradient enhances the trade winds over the equatorial waters. This results in a warm surface current that moves east to west at the equator. The western Pacific develops a thick, warm layer of water while the eastern Pacific has the cold Humboldt Current enhanced by upwelling. However, in other years the Southern Oscillation, for unknown reasons, swings in the opposite direction, dramatically changing the usual conditions described above, with pressure increasing in the western. Pacific and decreasing in the eastern Pacific. This change in the pressure gradient causes the trade winds to weaken or, in some cases, to reverse. This then causes the warm water in the western Pacific to flow eastward, increasing sea-surface temperatures in the central and eastern Pacific. The eastward shift signals the beginning of an El Nifio.

Scientists try to document as many past El Nino events as possible by piecing together bits of historical evidence, such as sea-surface temperature records, daily observations of atmospheric pressure and rainfall, fisheries' records from South America, and the writings of Spanish colonists dating back to the fifteenth century. From such historical evidence we know that El Nirtos have occurred as far back as records go. ■ It would seem that they are becoming more frequent. ■Records indicate that during the sixteenth century, an El Nino occurred on average every six years. ■ Evidence gathered over the past few decades indicates that El Ninos are now occurring on average a little over every two years. ■Even more alarming is the fact that they appear to be getting stronger. The 1997-1998 El Nifio brought copious and damaging rainfall to the southern United States, from California to Florida. Snowstorms in the northeast portion of the United States were more frequent and intense than in most years.

1. The word "approaches" in the passage is closest in meaning to
O nears
O crosses
O travels along
O leaves

2. According to paragraph 1, what happens when the Humboldt Current interacts with westward flowing trade winds?
O Anchovies from southern waters are carried northward.
O Cold water from lower depths is brought closer to the surface.
O The Humboldt Current stops flowing toward the equator.
O The Humboldt Current begins to flow closer to the coasts of Ecuador and Peru.

3.Which of the following questions about the El Nino phenomenon is NOT answered in paragraph 2 ?
O Why is the El Niflo phenomenon called El Nirto?
O How do fishers spend their time during the El Nifio season?
O How do coastal fish obtain enough nutrients during the El Nifio season?
O Is the temperature of coastal waters different during the El Niflo season than it is the rest of the year?

4. The word "exceptionally" , in the passage is closest in meaning to
O obviously
O unusually
O relatively
O occasionally

5. Paragraph 3 supports which of the following statements about El Ninos, as that term is

now used?
O El Nifios can originate in areas other than the Pacific Ocean.
O El Nifios can arise when warm currents last for two months or less.
O El Nifios affect water temperatures long distances from the South American coast.
O Multiple El Niflos can arise within a single calendar year.

6. The phrase "is usually accompanied by" in the passage is closest in meaning to
O usually develops before
O usually occurs together with
O is usually indicated by
O is usually caused by

7. The word "jointly" in the passage is closest in meaning to
O together
O therefore
O rightfully
O simply

8. According to paragraph 4, what did Sir Gilbert Walker discover?
O There is a close link between El Nifio and the Southern Oscillation.
O Surface pressure readings all across the Pacific first rise and then fall before an El Nifio occurs.
O Surface pressure on one side of the Pacific tends to fall when pressure rises on the opposite side.
O The formation of an El Nirto depends on conditions all across the Pacific, not just in the waters off of South America.

9. According to paragraph 5, what is the end result of the east-to-west pressure gradient in the eastern Pacific during a typical year?
O The formation of a thick, warm layer of water in the western Pacific
O The reversal of the pressure gradient to west-to-east by the end of the year
O A change in the direction of the Southern Oscillation
O The eastward flow of warm water from the western Pacific

10. According to paragraph 5, all of the following changes occur in the Pacific before an El Nifio begins EXCEPT:
O Pressure increases in the western Pacific and decreases in the eastern Pacific.
O The trade winds decrease in intensity or reverse in the direction.
O Surface temperatures increase in the central and eastern Pacific.
O Ocean currents speed up as they move eastward.

11. What can be inferred about El Ninos from the historical evidence mentioned in paragraph 6 ?
O They have often brought damaging weather to parts of the United States.
O They have been occurring since at least the fifteenth century.
O They occurred less frequently in the sixteenth century than in the fifteenth.
O They have had stronger weather effects on the United States in recent decades than on

other locations.

12. Why does the author include the information that in 1997-1998 "Snowstorms in the northeast portion of the United States were more frequent and intense than in most years"?
O To provide evidence supporting the claim that El Nifios are getting stronger
O To explain why the southern United States experienced copious and damaging rainfall in 1997-1998
O To show that traditional methods are not adequate for documenting the effects of El Niftos
O To identify a consequence of the fact that El Niflos are now occurring a little over once every two years

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

There is clear support for this view in the available documents.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your answer choices to the spaces where they belong To remove an answer choice, click on it.

To review the passage, click VIEW TEXT

As it is currently used, the term El Nirio refers to a strong and lengthy disruption to the normal pattern of ocean currents, water temperatures, and winds in the Pacific.

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Answer Choices

An El Nino typically begins when the Humboldt Current causes upwelling as it travels toward the equator along the coast of Peru and Ecuador.

El Nirlos are preceded by the reversal of the usual east-to- west pressure gradient in the Pacific, the weakening or reversal of the trade winds, and the movement of warm water eastward.

Comparisons of historical records with recent past events show that El Ninos are becoming more frequent and stronger.

In an El Nino, warm surface currents replace the Humboldt Current for many months, raising ocean temperatures far from the coast.

Scientists discovered the Southern Oscillation by taking surface-pressure readings at weather stations on both sides of the Pacific.

In recent decades, El Nifios have begun to occur north of the equator and thereby affect weather conditions in the United States.

[Listening]

Conversation 1

1.What do the speakers mainly discuss?

O The lack of functional printers in the student center

O The large crowds in the computer labs

O The skills that computer and printer technicians need

O The expected delivery of new computer printers

2. Why does the man mention the engineering school?

O To direct the student to a place where she can finish her work

O To indicate who serves as computer technicians

O To indicate where computer technicians are currently busy fixing printers

O To indicate that the problem with the printers is not limited to the student center

3. Why does the man mention the president of the college?

O To suggest that the president is too busy to get involved in this issue

O To indicate that the president proposed hiring additional technicians

O To explain that the president decided that purchasing new printers would be too costly
O To point out the president's involvement in acquiring new printers

4. What does the man offer to do?
O Replace empty ink cartridges in printers in the student center
O Allow the student to use computer centers that had been closed
O Send a technician to the student center to repair the printers
O Send an e-mail to the students to explain when the new printers will be installed

5. Why does the man say this: 
O To indicate that he does not understand the purchasing process
O To assure the student that the printers will be installed next month
O To suggest that the student should contact the purchasing office for further information
O To help explain the reason for the delay in receiving the new printers

Lecture 1
1. What is the main purpose of the lecture?
O To explain how the red pigment in leaves breaks down
O To show that leaf color varies based on the tree species
O To introduce a theory about why leaves turn a particular color
O To explain how chlorophyll protects trees in autumn

2.What does the professor imply when she explains why leaves are green?
O She wants to correct a common misconception about the topic.
O She thinks the students are probably already familiar with the material.
O She believes the process is too complicated to discuss in depth.
O She knows that students are often confused about the functions of chlorophyll

3. What does the professor mean when she says that the classic theory is partially right?
O It describes what happens in the summer but not what happens in autumn
O It describes what happens in tree leaves but not what happens in leaves of other plants
O It explains how pigments are synthesized but not how they break down.
O It explains some cases of color change in tree leaves but not all cases.

4. Why does the professor mention painting a car?
O To question why a large amount of anthocyanin is produced just before leaves fall
O To explain why most leaves turn red instead of other colors
O To remind students how cooler temperatures affect the color of leaves
O To show how anthocyanin absorbs sunlight to produce food for trees

5. The professor mentions theories about why leaves turn red that involve predatory insects and fungi. What is her opinion about those theories?
O They are based on careful research.
O They do not completely explain the phenomenon.
O They have not received enough attention.

O They have been proved to be incorrect.

6. According to the professor, why does anthocyanin appear on the upper side of some leaves?
O To help chlorophyll absorb the sunlight
O To maximize the leafs utilization of sunlight
O To accelerate the breakdown of chlorophyll
O To protect an important process from the sunlight

Lecture 2

1. What is the main purpose of the lecture?
O To explain a mechanism behind the ability to approximate numbers
O To explore the connection between ability in symbolic mathematics and the ability to approximate numbers
O To show the importance of new research into the ability to solve complex mathematical problems
O To demonstrate that children, adults, and animals have a similar ability to approximate numbers

2. Why does the professor mention six-month-old infants?
O To emphasize that ANS is largely innate
O To refute the claim that symbolic mathematics is learned
O To point out the difficulty of testing mathematics ability in very young children
O To contrast the way infants learn with how older children learn

3. Why does the professor stress that the dots in the experiment flashed on the computer screen for only a fraction of a second?
O To emphasize that humans' ANS ability is more developed than that of animals
O To point out that it was not possible to complete the task using formal mathematics
O To show a contrast between the dot experiment and the color-naming experiment
O To explain, in part, how subjects were chosen for the experiment

4. What did researchers observe in the study of fourteen-year-old children?
O The children with strong ANS skills also scored well on color-naming tests
O The children were more likely to make mistakes when there were small numbers of blue and yellow dots
O The ANS skills of the children had improved over time.
O There were large differences in the ANS skills of the children tested.

5. Why does the professor mention that the subjects of the experiment were also tested in reading and word knowledge?
O To show that ANS skills are not linked with abilities in those areas
O To emphasize the thoroughness of the researchers
O To point out that ANS and other skills are learned in a similar way
O To contrast learned skills with innate abilities

6. What is the professor's opinion of using instruction in ANS to improve children's performance in formal mathematics?

O It is likely that instruction in ANS would lead to improvement in areas other than formal mathematics.

O It would be important for the instruction in ANS to begin when children are very young.

O It is unclear whether instruction in ANS would improve performance in formal mathematics.

O it is more likely that instruction in formal mathematics would improve children's ANS ability.

Conversation 2

1. What are the speakers mainly discussing? (Click on 2 answers.)

□ he man's concerns about the midterm exam

□ An acting award that Professor Davis recently won

□ A professor's playwriting accomplishments

□ Arrangements for attending a local play

2. What does Professor Davis plan to give the student? (Click on 2 answers.)

□ Directions to a theater

□ A list of students' addresses

□ Tickets to a play

□ A study guide for the midterm exam

3. Why does Professor Davis want the students to attend a reception?

O So that the students can hear a talk about script adaptation

O So that the students can see a professor receive an award

O So that the students can ask an actor questions

O So that the students can meet the director of a play

4. What does the professor imply about script adaptation?

O It depends almost entirely on the writer's imagination.

O It is more often based on novels than on short stories.

O It is more difficult than the man thinks it is.

O It gives playwrights more commercial success than writing original plays does

5. Why does the professor say this:

O To express doubt about the quality of local plays

O To indicate that the man's assumption is wrong

O To encourage the man to see local productions

O To indicate that she agrees with the man

Lecture 3

1. What is the lecture mainly about?
O Early influences that shaped the career of Theodor Seuss Geisel
O The use of Dr. Seuss books in modern elementary schools
O The literary and artistic approach of Theodor Seuss Geisel
O Two prominent authors of twentieth-century children's literature

2. According to the professor, why did teachers oppose using Dr. Seuss books in the classroom during the 1950s and 1960s?
O Teachers thought the books were boring
O Teachers associated the books with play rather than schoolwork
O Dr. Seuss books used vocabulary that was not on the Dolch list.
O Dr. Seuss books could not be used to teach subjects other than reading.

3. Why does the professor mention the citation awarded to Geisel by the Pulitzer Prize Committee?
O To emphasize how long it took for Geisel，s literary contributions to be appreciated
O To emphasize the difficulty of writing books that appeal to both children and adults
O To explain how authors of children's literature were typically honored
O To explain why Geisel's books finally became popular

4. What does the professor say about Geisel's work as an illustrator?(Click on 2 answers) □ Geisel's approach to drawing scenery is more sophisticated than it first appears
□ Geisel's style was strongly influenced by earlier illustrators of children's books.
□ Geisel's human characters all look very much alike.
□ Geisel's style is widely taught in art schools today.

5. What was the connection between Geisel and John Hersey?
O Their writing styles were remarkably similar.
O They collaborated on an article about teaching children to read
O Hersey's article inspired Geisel to write a new kind of book.
O Hersey wrote a novel that was inspired by Geisel's career.

6. What is the professor's opinion of Geisel's book The Cat in the Hat?
O It is effective because its characters are people and animals rather than imaginary creatures
O it would be a better teaching tool if it had more challenging vocabulary.
O It wrongly encourages children to break their parents' rules.
O It cleverly presents moral issues in an entertaining way.

Lecture 4

1. What does the professor mainly discuss?
O Methods of converting radio waves into sound waves
O Features of different types of electromagnetic radiation
O The various paths that very-low-frequency waves follow on Earth
O The emission and detection of very-low-frequency waves

2. What is one difference between radio waves and sound waves that the professor emphasizes?
O Radio waves have a lower frequency.
O Water stops radio waves from spreading but does not stop sound waves
O Unlike sound waves, radio waves can travel outside Earth's atmosphere.
O Naturally occurring radio waves are difficult to detect on Earth at night.

3. What explanation does the professor give for the constant occurrence of VLF emissions on Earth?
O At any given time, some part of the world is experiencing sunrise or sunset
O Waveguides constantly form in the atmosphere.
O Earth’s magnetosphere directs interplanetary waves toward Earth’s surface.
O Lightning occurs constantly on the planet.

4. Why are sunrise and sunset the best times to listen to VLF signals?
O Because thunderstorms are most likely to occur then
O Because radio waves travel through natural waveguides then
O Because higher-frequency signals are less active then
O Because temperatures are not extremely high or low then

5. Why does the professor discuss whistlers and tweeks?
O To illustrate that the path a VLF wave travels can affect the sound it makes on a radio
O To point out that VLF waves can affect the sounds heard on a household or car radio
O To describe how a colleague discovered the origin of VLF waves
O To clarify the difference between VLF waves and other kinds of waves

6. What does the professor imply when he says this:
O He needs to think before he can answer the woman's question.
O The woman has underestimated how often VLF waves can be detected.
O The woman does not realize that waiting for a thunderstorm can take a long time.
O The woman does not understand the relationship between thunderstorms and lightning.

[Speaking]

1. People set a variety of goals for themselves throughout their lives. Describe one goal you would like to achieve in the future, and explain why this goal is important to you. Include specific details in your explanation.

2. Some students attend college full-time, while others attend college part-time. Which do you think is better? Explain why.

3. *Reading Time:*
University Makes Changes to Orientation Program
Madison University is making a change to the orientation program for first-year students. In the past, as part of orientation, new, incoming students could go on a two-day hiking and camping trip together with other incoming students on the weekend before classes begin. In order to encourage more students to take advantage of the opportunity to get to know one another in an informal setting, the university will now offer a choice of activities: students will be able to either go hiking or participate in organized group games on campus. Additionally, these activities will last one day only, not two, a change many students had requested.

The man expresses his opinion about the change described in the article. Briefly summarize the change Then state his opinion about the change and explain the reasons he gives for holding that opinion.

4. *Reading Time:*
Population Changes
Populations of living beings are constantly changing The number of humans, animals, insects, or plants living in a given area can vary because of two kinds of factors: biotic and abiotic Biotic factors are living factors that can influence the size of populations, such as predators or other species competing for food. Abiotic factors are nonliving things in the surrounding environment that can cause population changes, such as weather or sunlight. Biotic and abiotic factors cause continual changes in the number of individuals that make up a population of organisms.

Using the examples of mice and rabbits from the lecture, describe the two different types of factors that can cause population changes.

5.

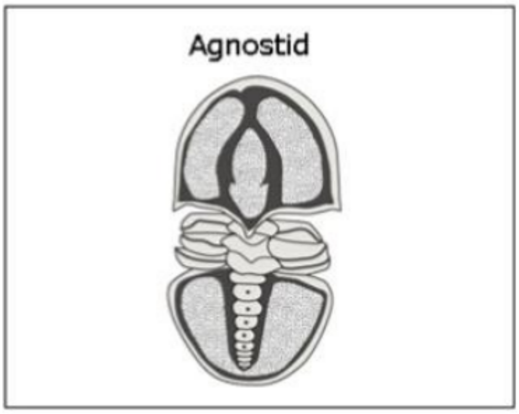
The speakers discuss two possible solutions to their problem. Briefly summarize the problem Then state which solution you prefer and explain why.

6.

The speakers discuss two possible solutions to their problem Briefly summarize the problem Then state which solution you prefer and explain why.

[Writing]

1.(reading)



Agnostids were a group of marine animals that became extinct about 450 million years ago. Agnostid fossils can be found in rocks in many areas around the world. From the fossil remains, we know that agnostids were primitive arthropods-relatives of modern-day insects. However, the fossil information does not allow paleontologists to determine with certainty what agnostids ate or how they behaved. There are several different theories about how agnostids may have lived.

Free-Swimming Predators

First, the agnostids may have been free-swimming predators that hunted smaller animals. It is known that other types of primitive arthropods were strong swimmers and active predators, so it is reasonable that the agnostids may have lived that way as well And while the agnostids were small, sometimes just six millimeters long, there were plenty of smaller organisms in the ancient ocean for them to prey on.

Seafloor Dwellers

Second, they may have dwelled on the seafloor. Again, there are examples of other types of primitive arthropods living this way, so it is possible that agnostids did too. On the seafloor they would have survived by scavenging dead organisms or by grazing on bacteria.

Parasites

Third, there is the possibility that the agnostids were parasites, living on and feeding off larger organisms. One reason that this seems possible is that there are many species of modern-day arthropods that exist as parasites, such as fleas, ticks, and mites. The agnostids might have lived on primitive fish or even on other, larger arthropods.

Directions: You have 20 minutes to plan and write your response. Your response will be judged on the basis of the quality of your writing and on how well your response presents the points in the lecture and their relationship to the reading passage. Typically, an effective response will be 150 to 225 words.

Question: Summarize the points made in the lecture, being sure to explain how they challenge the specific theories presented in the reading passage.

Directions: Read the question below You have 30 minutes to plan, write, and revise your essay. Typically, an effective response will contain a minimum of 300 words.

Question:

Imagine that you are in a classroom or a meeting. The teacher or the meeting leader says something incorrect In your opinion, which of the following is the best thing to do?

-Interrupt and correct the mistake right away

-Wait until the class or meeting is over and the people are gone, and then talk to the teacher or meeting leade

-Say nothing

Use specific reasons and examples to support your answer.